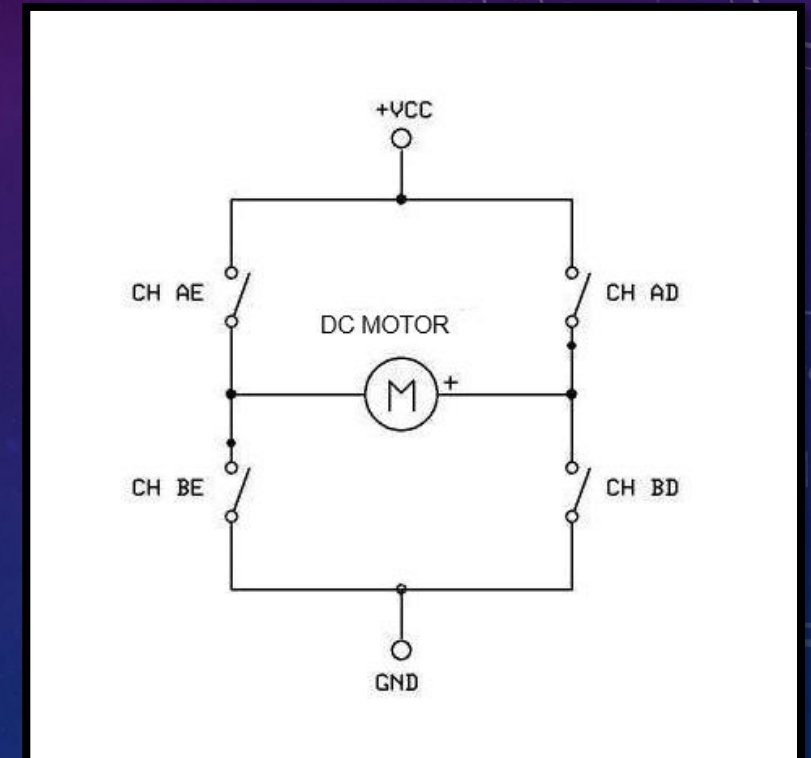


The background features a dark blue gradient with faint, light blue concentric circles and degree markings (40, 150, 160, 170, 180, 190, 200, 210, 220, 230, 240, 250, 260) on the left side, suggesting a technical or scientific theme.

RELAY-BASED H-BRIDGE WITH TRANSISTOR DRIVERS, ARDUINO CONTROL AND LCD INTERFACE

H-BRIDGE CIRCUIT OVERVIEW

An **H-Bridge** is an electronic topology that enables a DC motor to be driven in both directions. By using four switching elements—**relays** in this specific project—it allows the reversal of voltage polarity across the motor terminals. When diagonal pairs of switches are closed, the motor spins either clockwise or counter-clockwise, while opening all switches brings the motor to a safe stop. This custom shield implementation ensures that high-current loads (up to 10 A) are managed efficiently and safely by the Arduino controller.



Manufacturing & Validation

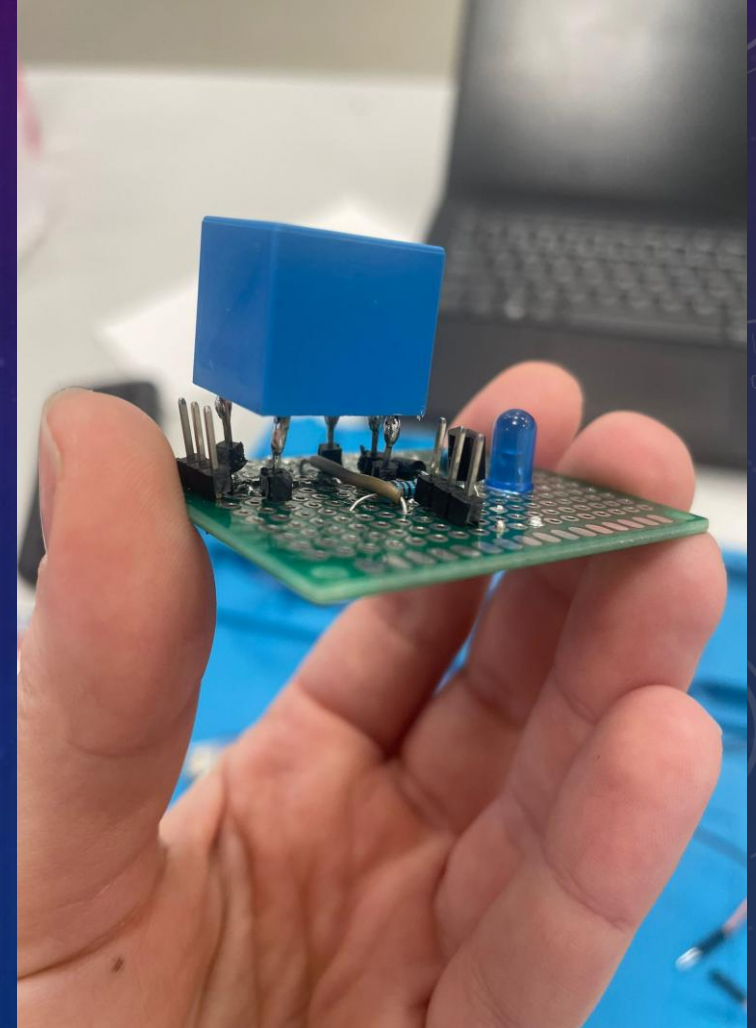
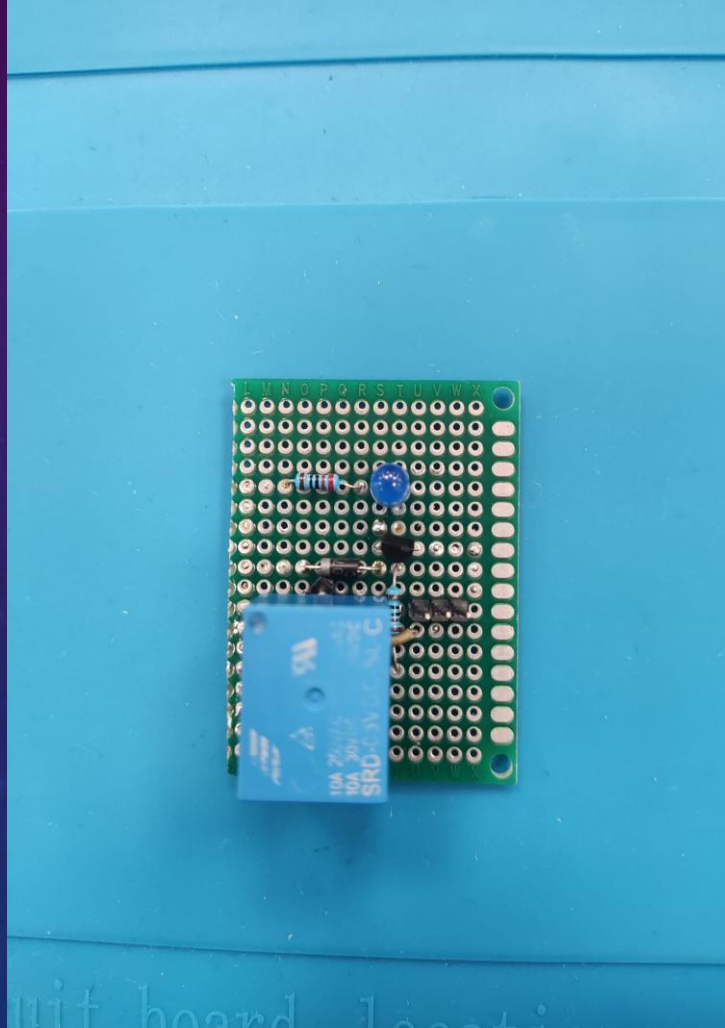
Custom Build: Physical transition from EDA design to a functional single-sided PCB shield.

Switching Logic: 2N3904 transistors with 1k resistors for reliable relay activation.

Visual Alerts: Integrated LEDs to indicate real-time motor direction and switching status.

Circuit Safety: 1N4007 freewheel diodes to protect the Arduino from inductive spikes.

Build Quality: Industrial-grade THT soldering (proper wetting and fillets) for durability.



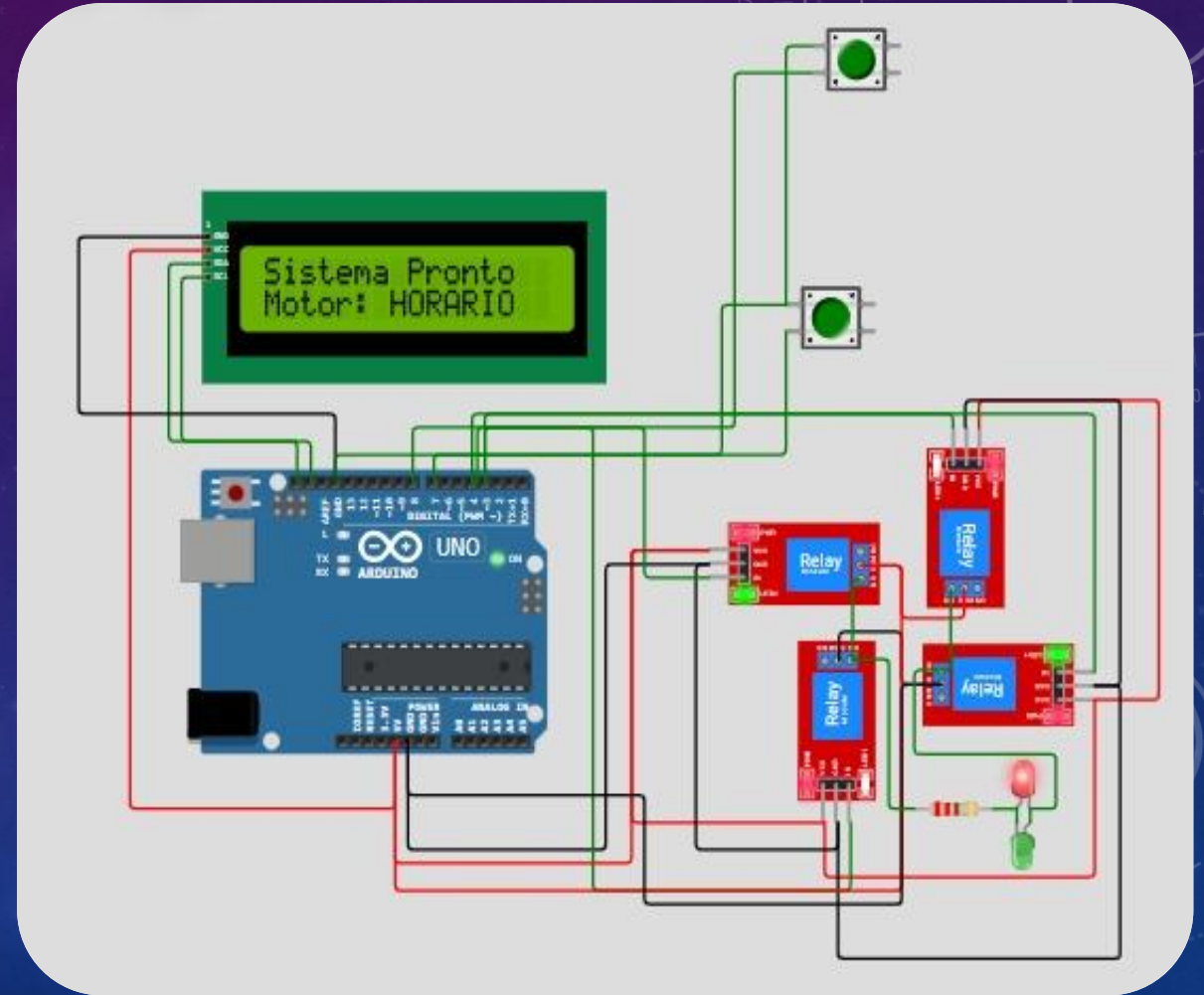
Simulation Software

Control Unit (Arduino Uno): Acts as the processing core, reading user inputs and managing switching timings to prevent short-circuits (shoot-through).

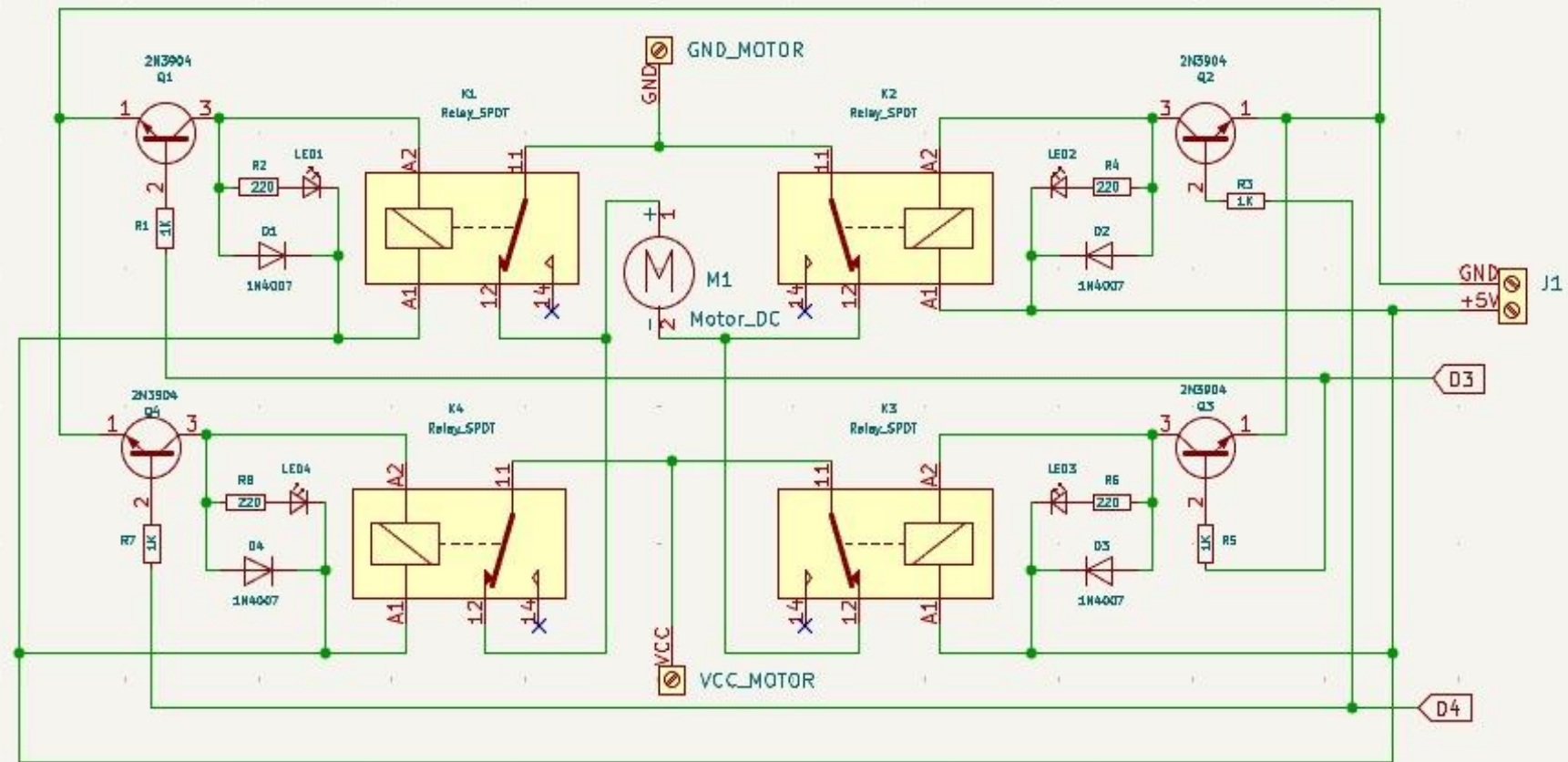
User Interface (UI): Comprised of two buttons configured with internal pull-up resistors and a 16x2 LCD screen that provides instant feedback on rotation direction and system status.

Power Stage (Relay H-Bridge): The core of this project is the creation of a custom shield housing for four relay modules. This switches that reverse the voltage polarity sent to the motor, enabling bidirectional control.

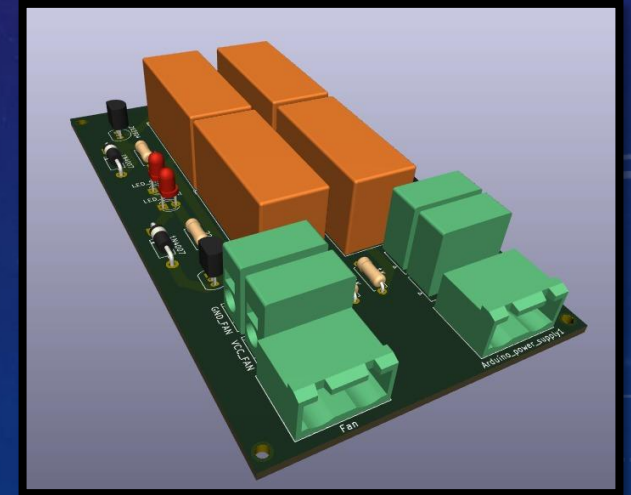
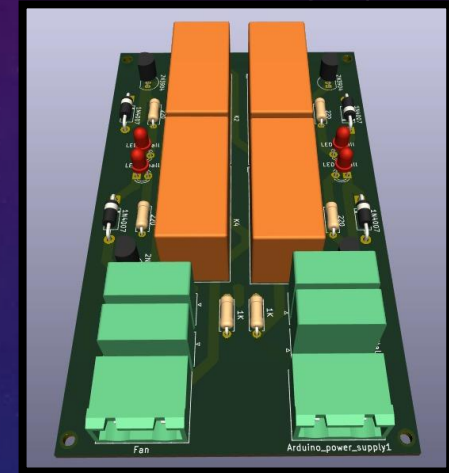
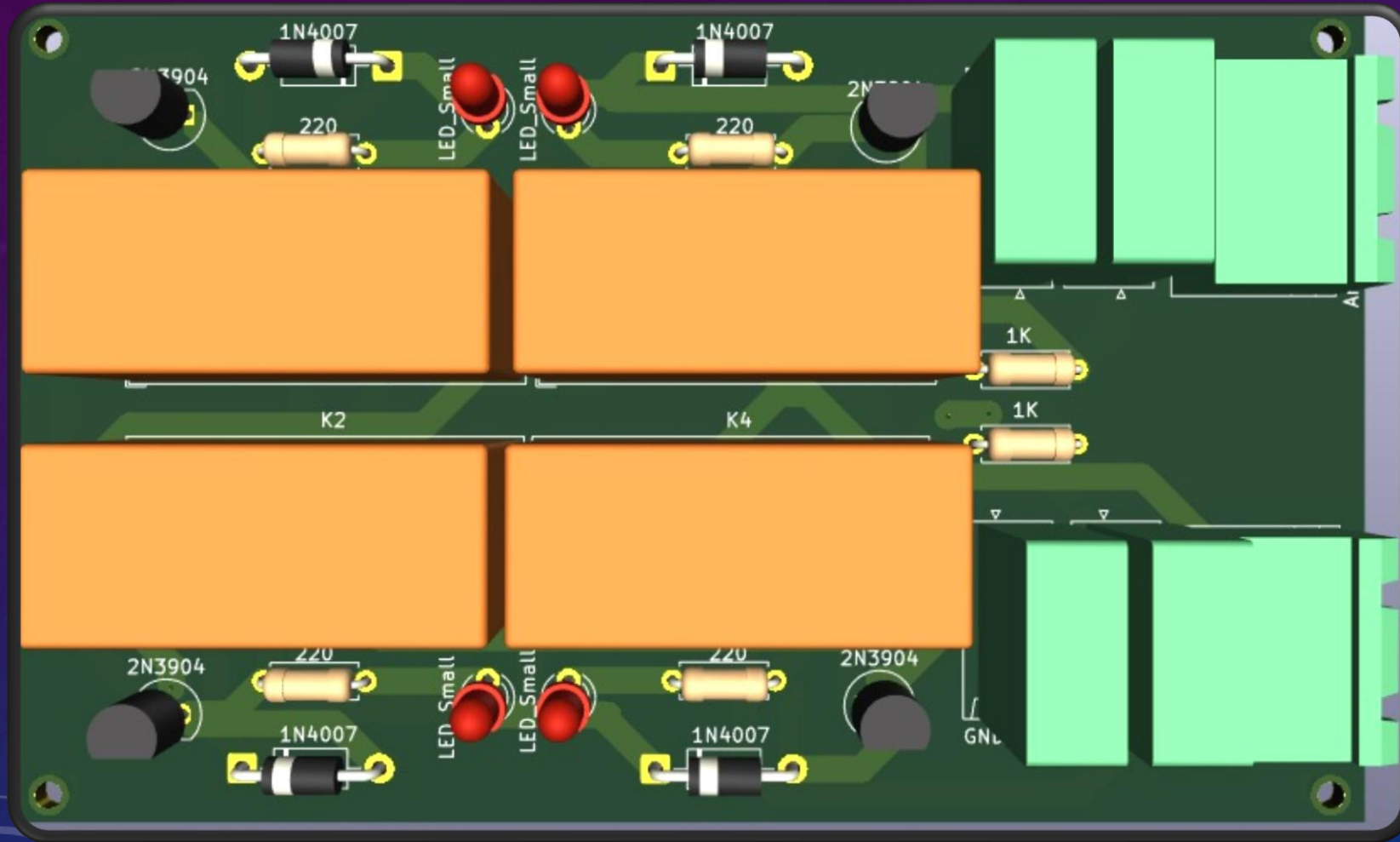
Signal Conditioning: Each relay is activated by an NPN transistor that amplifies the low-current signal from the Arduino to the current level required these relays to act as physical to energize the relay coil.



Electronic Schematic



Finished PCB



Hardware Demonstration

